

Nonconventional Energy sources (BT EE 707)



हरियाणा केंद्रीय विश्वविद्यालय
CENTRAL UNIVERSITY OF HARYANA
NAAC ACCREDITED 'A' GRADE UNIVERSITY

Unit 2: Solar Power

- SOLAR POWER OVERVIEW
- PV TECHNOLOGY BASICS

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PV technology basics

How PV Cells Work

A typical silicon PV cell is composed of a thin wafer consisting of an ultra-thin layer of phosphorus-doped (N-type) silicon on top of a thicker layer of boron-doped (P-type) silicon.

An electrical field is created near the top surface of the cell where these two materials are in contact, called the P-N junction.

When sunlight strikes the surface of a PV cell, this electrical field provides momentum and direction to light-stimulated electrons, resulting in a flow of current when the solar cell is connected to an electrical load

Solar cells are semiconductor devices

that produce electricity from sunlight via the photovoltaic effect.

Sunlight strikes the cell, photons with energy above the semiconductor bandgap impart enough energy to create electron-hole pairs.

A junction between dissimilarly doped semiconductor layers sets up a potential barrier in the cell, which separates the light-generated charge carriers.

This separation induces a fixed electric current and voltage in the device. The electricity is collected and transported by metallic contacts on the top and bottom surfaces of the cell.

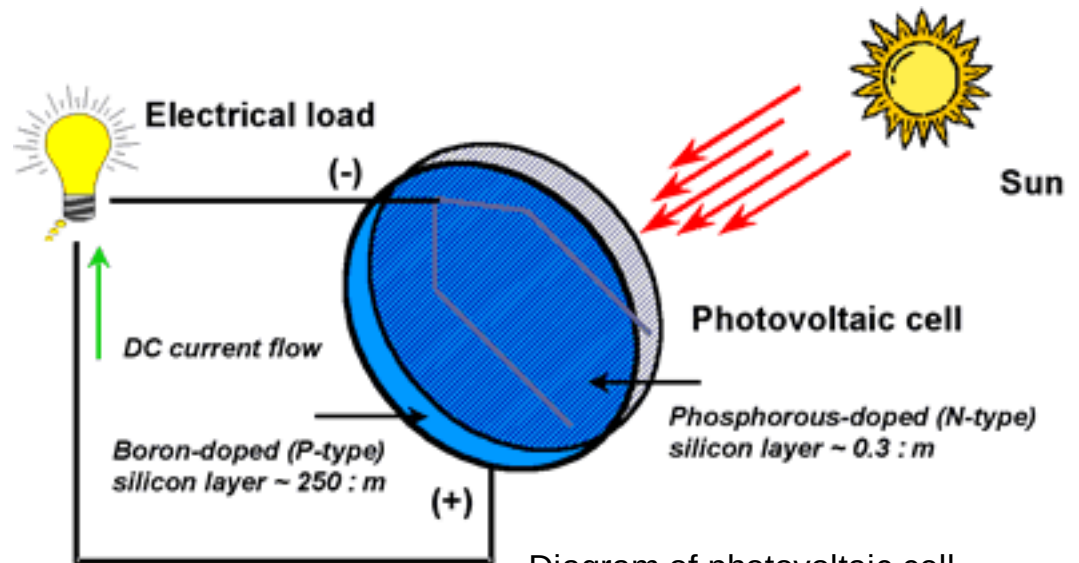
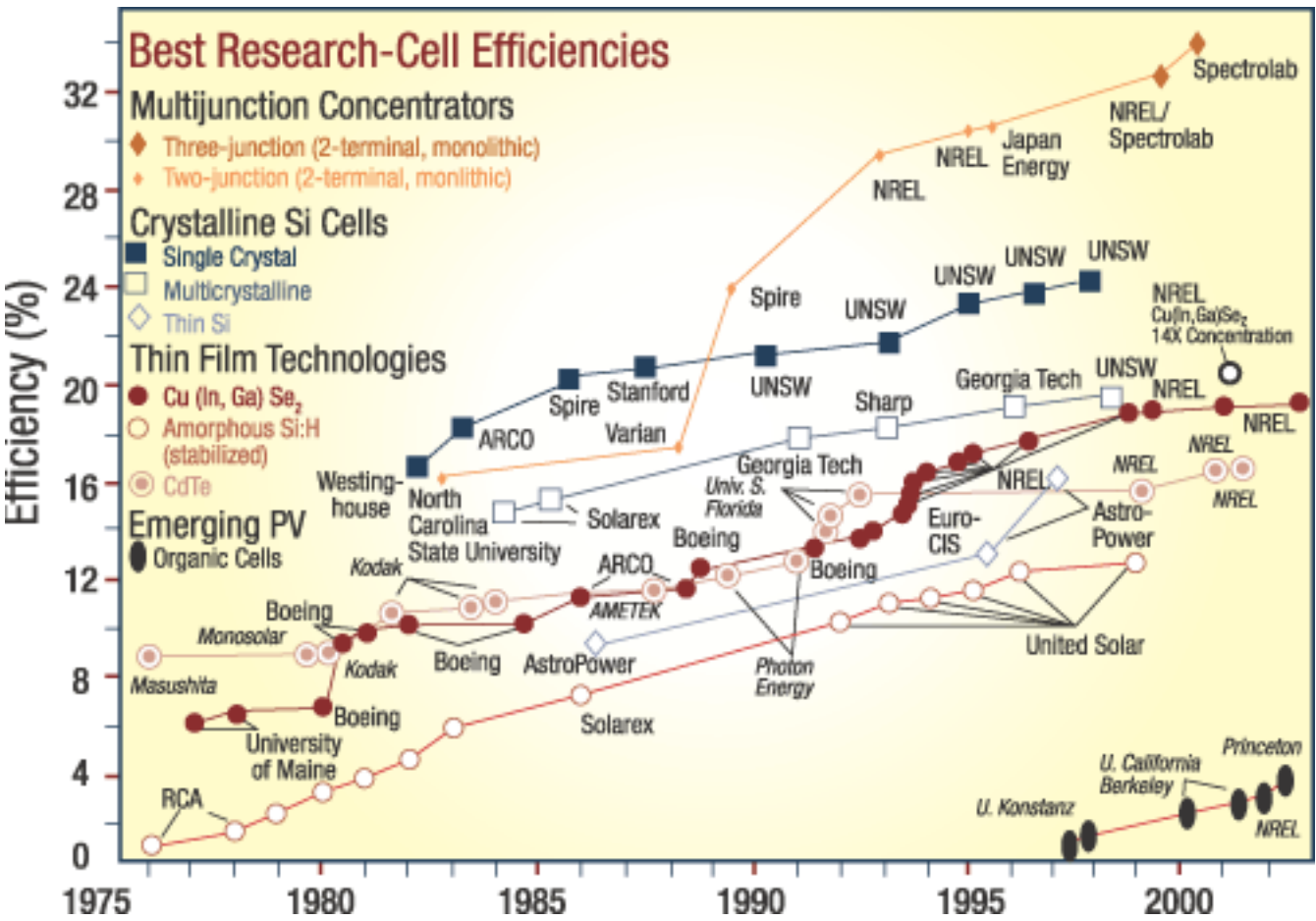


Diagram of photovoltaic cell.

PV technology basics

Applicable semiconductor materials> Cell Efficiencies

- silicon (Si),
- gallium arsenide (GaAs),
- cadmium telluride (CdTe),
- copper indium diselenide (CIS),
- hydrogenated amorphous silicon.



There are significant differences between the best performance and the theoretically predicted value for each material; closing these gaps is the subject of ongoing research (see figure).

PV technology basics

In the case of a single-junction device, the efficiency of the solar cell, the ratio of the power produced, and the incident light power are limited.

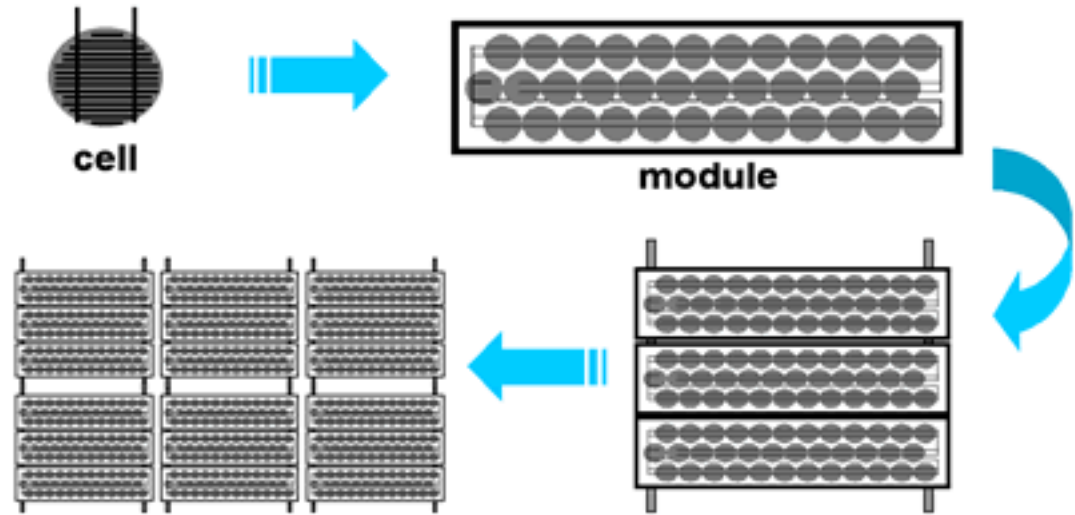
Photons with energies below the bandgap of the material produce only heat. Excess energy above that needed to generate electron-hole pairs also produces heat.

A multijunction device, in which two or more solar cells are stacked on top of each other, can exploit different portions of the solar spectrum.

For example, a four-junction device with bandgaps of 1.8, 1.4, 1.0, and 0.7 electron volts (eV) results in a theoretical efficiency of more than 52%.

The multijunction approach, however, presents significant challenges in both materials preparation and device design.

PV Cells, Modules, & Arrays



Photovoltaic cells are connected electrically in series and/or parallel circuits to produce higher voltages, currents and power levels.

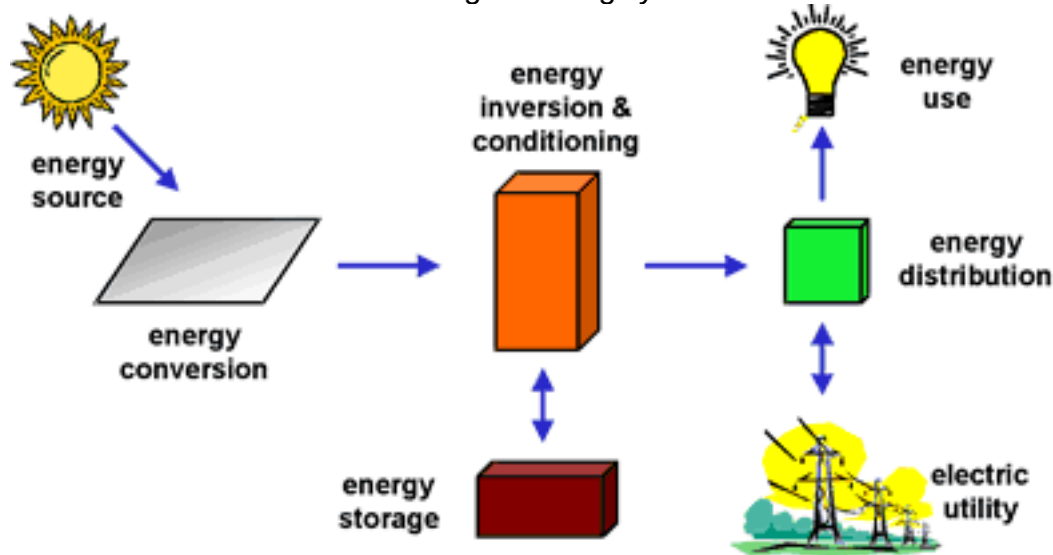
Photovoltaic modules consist of PV cell circuits sealed in an environmentally protective laminate, and are the fundamental building block of PV systems.

Photovoltaic panels include one or more PV modules assembled as a pre-wired, field-installable unit. A photovoltaic array is the complete power-generating unit, consisting of any number of PV modules and panels.

PV technology basics

How a PV System Works

PV systems are like any other electrical power generating systems, just the equipment used is different than that used for conventional electromechanical generating systems.



Depending on the functional and operational requirements of the system, the specific components required, and may include major components

**DC-AC power inverter,
battery bank,
system and battery controller,
auxiliary energy sources**

and sometimes the **specified electrical load (appliances)**.

In addition, an assortment of balance of system (BOS) hardware,

Including wiring, overcurrent, surge protection and disconnect devices, and other power processing equipment.

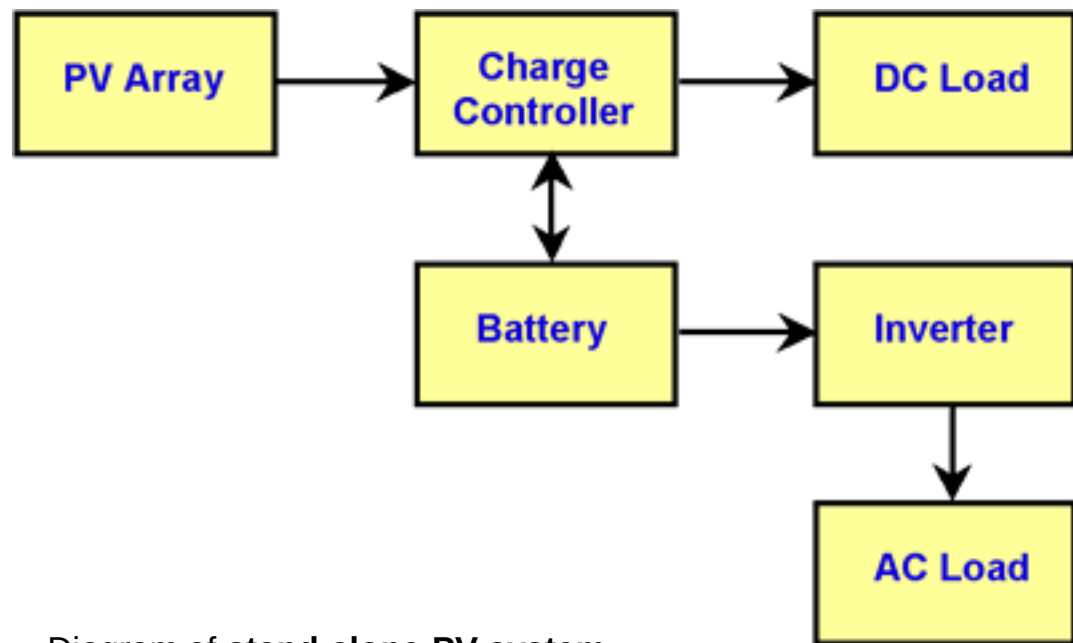
PV technology basics

Why Are Batteries Used in Some PV Systems?

Batteries are often used in PV systems for the purpose of storing energy produced by the PV array during the day, and to supply it to electrical loads as needed (during the night and periods of cloudy weather).

Other reasons batteries are used in PV systems are to operate the PV array near its maximum power point, to power electrical loads at stable voltages, and to supply surge currents to electrical loads and inverters.

In most cases, a battery charge controller is used in these systems to protect the battery from overcharge and overdischarge.



In many stand-alone PV systems, batteries are used for energy storage. Figure shows a diagram of a typical stand-alone PV system powering DC and AC loads

Diagram of **stand-alone PV system** with battery storage powering DC and AC loads.

PV technology basics

Types of PV Systems

How Are Photovoltaic Systems Classified?

Photovoltaic power systems are generally classified according to:

- functional and operational requirements,
- component configurations,
- how the equipment is connected to other power sources and electrical loads.

The two principle classifications are
grid-connected or utility-interactive systems
stand-alone systems.

Photovoltaic systems can be designed to provide DC and/or AC power service, can operate interconnected with or independent of the utility grid, and can be connected with other energy sources and energy storage systems.

1.7.1 Grid-Connected (Utility-Interactive) PV Systems.

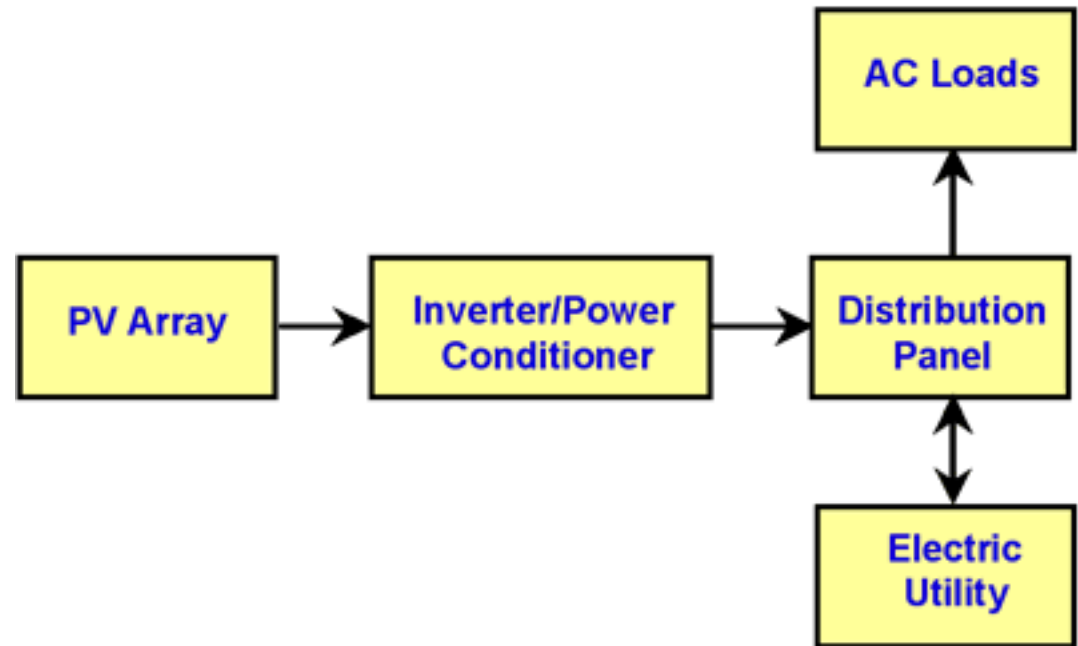


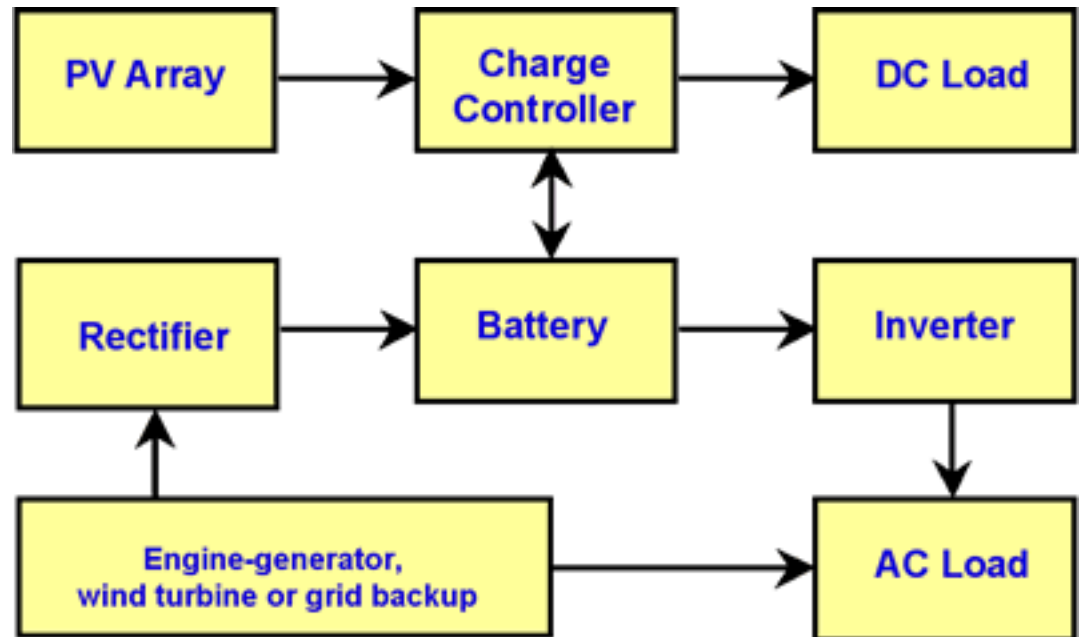
Diagram of **grid-connected photovoltaic system**

PV technology basics

Stand-alone PV systems are designed to operate independent of the electric utility grid, and are generally designed and sized to supply certain DC and/or AC electrical loads.

These types of systems may be powered by a PV array only, or may use wind, an engine-generator or utility power as an auxiliary power source in what is called a PV-hybrid system.

photovoltaic hybrid system.



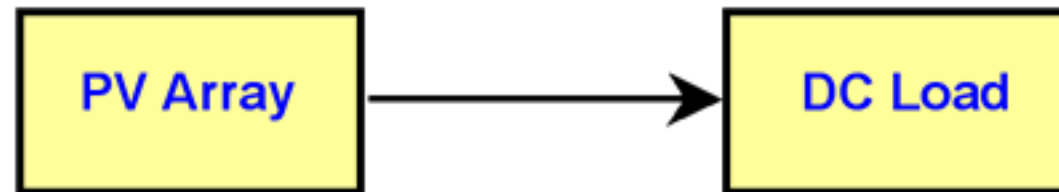
PV technology basics

The simplest type of stand-alone PV system is a direct-coupled system, where the DC output of a PV module or array is directly connected to a DC load

Since there is no electrical energy storage (batteries) in direct-coupled systems, the load only operates during sunlight hours, making these designs suitable for common applications such as ventilation fans, water pumps, and small circulation pumps for solar thermal water heating systems.

Matching the impedance of the electrical load to the maximum power output of the PV array is a critical part of designing well-performing direct-coupled system.

For certain loads such as positive-displacement water pumps, a type of electronic DC-DC converter, called a maximum power point tracker (MPPT) is used between the array and load to help better utilize the available array maximum power output.



Direct-coupled PV system.

PV technology basics

crystalline silicon

The conventional technology that supports more than **85% of today's PV market is based on wafers of crystalline Si.**

Production capacities are expected to more than double in the next three to five years, and the products are proven and accepted in the markets.

Single-crystal ingots are pulled from the melt, or polycrystalline ingots are cast in a crucible that is consumed in the process. Technical advances include the growth of ingots as big as 300 kg, the growth of multiple ingots with melt replenishment, and the reduction of consumable materials and energy costs. Significant R&D effort has been focused on reducing defects, improving doping, and automating the growth process. **Still, for single-crystal Si, the best commercial module efficiencies are only about 15%.**

How PV Cells Are Made

The process of fabricating conventional single- and polycrystalline silicon PV cells begins very pure semiconductor-grade polysilicon - a material processed from quartz and used extensively throughout the electronics industry.

The polysilicon is then heated to melting temperature, and trace amounts of boron are added to the melt to create a P-type semiconductor material.

Next, an ingot, or block of silicon is formed, commonly using one of two methods:

- 1) by growing a pure crystalline silicon ingot from a seed crystal drawn from the molten polysilicon
- or 2) by casting the molten polysilicon in a block, creating a polycrystalline silicon material.

Individual wafers are then sliced from the ingots using wire saws and then subjected to a surface etching process. After the wafers are cleaned, they are placed in a phosphorus diffusion furnace, creating a thin N-type semiconductor layer around the entire outer surface of the cell.

Next, an anti-reflective coating is applied to the top surface of the cell, and electrical contacts are imprinted on the top (negative) surface of the cell.

An aluminized conductive material is deposited on the back (positive) surface of each cell, restoring the P-type properties of the back surface by displacing the diffused phosphorus layer.

Each cell is then electrically tested, sorted based on current output, and electrically connected to other cells to form cell circuits for assembly in PV modules.



Image: A solar cell, made from a poly-crystalline silicon wafer

PV technology basics

Si ribbon and sheet technologies, which avoid the cost and material losses associated with slicing ingots, are the first of the new PV technologies to be commercialized. About 6% of today's PV modules are made using these methods. Most notable are the edge-defined, film-fed growth process, the string ribbon process, and the Si-film process (growth of a high-speed, continuous Si sheet of a variety of substrates). **Cell and module efficiencies are similar to those for polycrystalline Si wafers from ingots—about 12% at best—with somewhat lower efficiencies for the small-grained sheet materials.**

Ongoing incremental improvements will probably reduce manufacturing costs and increase module performance, keeping these technologies as the mainstay of PV markets for years to come.

Thin-Film Photovoltaics

Thin-film photovoltaic modules are manufactured by depositing ultra-thin layers of semiconductor material on a glass or thin stainless-steel substrate in a vacuum chamber.

A laser scribing process is used to separate and weld the electrical connections between individual cells in a module.

Thin-film photovoltaic materials offer great promise for reducing the materials requirements and manufacturing costs for PV modules and systems.



PV technology basics

Pros and Cons of PV

Photovoltaic systems have a number of merits and unique advantages over conventional power-generating technologies.

PV systems can be designed for a variety of applications and operational requirements, and can be used for either centralized or distributed power generation.

PV systems have no moving parts, **are modular, easily expandable and even transportable in some cases. Energy independence and environmental compatibility are two attractive features of PV systems.**

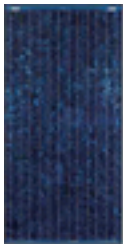
The fuel (sunlight) is free, and no noise or pollution is created from operating PV systems. In general, PV systems that are well designed and properly installed require minimal maintenance and have long service lifetimes.

At present, the **high cost of PV modules** and equipment (as compared to conventional energy sources) is the **primary limiting factor for the technology**. Consequently, the economic value of PV systems is realized over many years. In some cases, the surface area requirements for PV arrays may be a limiting factor. Due to the diffuse nature of sunlight and the existing sunlight to electrical energy conversion efficiencies of photovoltaic devices, surface area requirements for PV array installations are on the order of 8 to 12 m² (86 to 129 ft²) per kilowatt of installed peak array capacity.

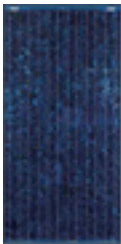
PV solar panel specifications



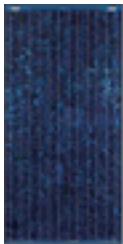
Most expensive
170 watt- 999.00



170 watt-783.00



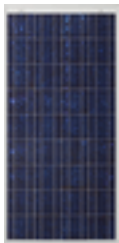
160 watt 799.



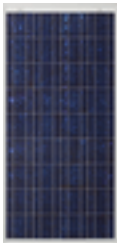
160 watt 725.



150 watt 699.



125 watt 629.00



125 watt 599.



80 watt 520.



65 watt 457.



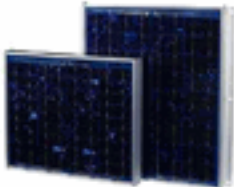
60 watt 462.00



50Watt=354.00



40watt+353.00



30Watt 259.00

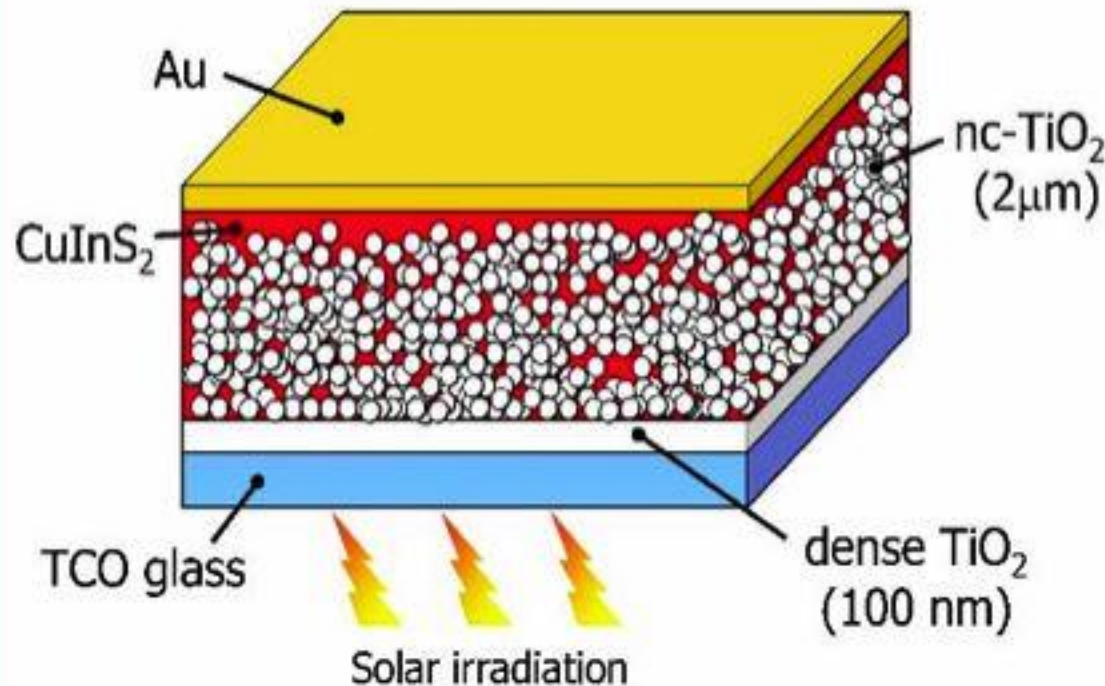


20watt 234.00



Cheapest
10 watt Solar panel- 149.00

3D solar cells



- 1 percolation of electrons and holes**
- 2 selective contacts**
- 3 no electrical shunts (pinholes)**
- 4 stoichiometric TiO₂**
- 5 p-type CuInS₂**
- 6 buffer layer**
- 7 intimate contact**
- 8 complete filling of pores**

M. Nanu, J. Schoonman, and A. Goossens, Advanced Materials 16 (2004) 453

M. Nanu, J. Schoonman, and A. Goossens, Adv. Func. Mat. 15 (2005) 95